Linear System Formulation and Dirichlet Boundary Conditions

In implicit finite difference formulations for the one-dimensional transient diffusion equation, the spatial coupling of unknown nodal values at the advanced time level $n+1$ leads to a coupled system of linear algebraic equations. This section presents the general coefficient matrix assembled for the interior nodes and establishes how Dirichlet boundary conditions are integrated into the algebraic framework.

A. System of Equations for Interior Nodes

Consider a one-dimensional domain discretized into N grid points indexed by $j = 0, 1, \ldots, N-1$. The physical boundaries are located at $j = 0$ (lower wall) and $j = N-1$ (upper wall). Consequently, the interior computational domain consists of $N-2$ unknown nodes:

$$j = 1, 2, \ldots, N-2$$

Evaluating an implicit discretization stencil at each interior node couples the unknown velocity u_j^{n+1} to its adjacent spatial neighbors u_{j-1}^{n+1} and u_{j+1}^{n+1}. For instance, employing the standard fully implicit (Laasonen) scheme yields the following discrete equation at node j:

$$-d\, u_{j-1}^{n+1} + (1+2d)\, u_j^{n+1} - d\, u_{j+1}^{n+1} = u_j^n \tag{1}$$

where the dimensionless diffusion number d is defined as:

$$d = \frac{\nu \Delta t}{(\Delta y)^2} \tag{2}$$

Writing Equation (1) for all interior nodes $j = 1, 2, \ldots, N-2$ simultaneously produces an $(N-2) \times (N-2)$ linear system of equations:

$$\begin{bmatrix} 1+2d & -d & 0 & \cdots & 0 \\ -d & 1+2d & -d & \ddots & \vdots \\ 0 & -d & 1+2d & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -d \\ 0 & \cdots & 0 & -d & 1+2d \end{bmatrix} \begin{bmatrix} u_1^{n+1} \\ u_2^{n+1} \\ u_3^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \end{bmatrix} = \begin{bmatrix} u_1^n \\ u_2^n \\ u_3^n \\ \vdots \\ u_{N-2}^n \end{bmatrix} + \begin{bmatrix} d\, u_0^{n+1} \\ 0 \\ \vdots \\ 0 \\ d\, u_{N-1}^{n+1} \end{bmatrix} \tag{3}$$

In Equation (3), the boundary contributions u_0^{n+1} and u_{N-1}^{n+1} appear directly on the right-hand side because their values are prescribed by the boundary conditions.

B. Incorporation of Dirichlet Boundary Conditions

For the physical Couette flow problem, the boundary values at the lower and upper boundaries are strictly prescribed by constant Dirichlet boundary conditions:

$$u_0^{n+1} = U_{\text{bottom}}, \qquad u_{N-1}^{n+1} = U_{\text{top}} \tag{4}$$

There are two primary approaches for incorporating these boundary conditions into the coefficient matrix and algebraic system.

Approach A: Reduced Interior System ($(N-2) \times (N-2)$)

In the reduced interior formulation, the known boundary values are excluded from the unknown solution vector. As shown in Equation (3), their contributions are absorbed directly into the right-hand side vector:

$$\mathbf{A}_{\text{int}}\, \mathbf{u}_{\text{int}}^{n+1} = \mathbf{b}_{\text{int}}^{n} \tag{5}$$

where $\mathbf{u}_{\text{int}}^{n+1} = [u_1^{n+1}, u_2^{n+1}, \dots, u_{N-2}^{n+1}]^T$ is of dimension $N-2$, and the right-hand side vector elements are given by:

$$\begin{aligned} b_1 &= u_1^n + d\, U_{\text{bottom}} \\ b_j &= u_j^n \quad \text{for } j = 2, 3, \dots, N-3 \\ b_{N-2} &= u_{N-2}^n + d\, U_{\text{top}} \end{aligned} \tag{6}$$

This approach minimizes the algebraic system to $(N-2)$ equations, but it introduces an index offset between the algebraic vector index (from 0 to $N-3$) and the physical spatial grid index (from 1 to $N-2$).

Approach B: Global Formulation ($N \times N$)

In the global system formulation, all computational nodes across the domain—including the boundary nodes—are preserved in the solution vector $\mathbf{u}^{n+1} \in \mathbb{R}^N$.

To close the system, the first and last rows of the coefficient matrix are formulated as identity equations corresponding to the Dirichlet boundary conditions in Equation (4):

$$\begin{aligned} \text{Row } 0: &\quad 1 \cdot u_0^{n+1} + 0 \cdot u_1^{n+1} + \cdots + 0 \cdot u_{N-1}^{n+1} = U_{\text{bottom}} \\ \text{Row } N-1: &\quad 0 \cdot u_0^{n+1} + \cdots + 0 \cdot u_{N-2}^{n+1} + 1 \cdot u_{N-1}^{n+1} = U_{\text{top}} \end{aligned} \tag{7}$$

Assembling the complete $N \times N$ global system yields:

$$
\begin{bmatrix}
1 & 0 & 0 & \cdots & 0 & 0 \\
-d & 1+2d & -d & \cdots & 0 & 0 \\
0 & -d & 1+2d & \ddots & \vdots & \vdots \\
\vdots & \ddots & \ddots & \ddots & -d & 0 \\
0 & \cdots & 0 & -d & 1+2d & -d \\
0 & 0 & \cdots & 0 & 0 & 1
\end{bmatrix}
\begin{bmatrix}
u_0^{n+1} \\ u_1^{n+1} \\ u_2^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \\ u_{N-1}^{n+1}
\end{bmatrix}
=
\begin{bmatrix}
U_{\text{bottom}} \\ u_1^n \\ u_2^n \\ \vdots \\ u_{N-2}^n \\ U_{\text{top}}
\end{bmatrix}
\tag{8}
$$

C. Selection for C++ Implementation

While Equation (5) and Equation (8) are mathematically equivalent for all interior nodes, the global $N \times N$ formulation defined in Equation (8) is adopted in the current C++ solver architecture.

The primary advantages of this formulation are:

1. **Uniform Indexing:** The row index of the algebraic system corresponds directly to the physical grid index j, which eliminates off-by-one indexing errors.
2. **Simplified Memory Layout:** The solution vector contains the complete domain profile, eliminating the need to manually reinsert boundary values after each linear solve.
3. **Boundary Condition Extensibility:** Alternative boundary conditions (such as Neumann or Robin boundaries) can be implemented simply by adjusting the discrete coefficients in row 0 or row $N-1$, without altering the dimension or structural mapping of the linear solver.